

CARDIAC OUTPUTS AND SHUNTS

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We take for granted the relative ease that we are now able to measure the rate at which the heart pumps blood, better known as the cardiac output. It is a central parameter for clinically important measurements such as the determination of the status of the patient in known or suspected shock and the calculation of valve area and vascular resistances (pulmonary and systemic vascular resistance). Because of the importance of cardiac output measurement, it is necessary to understand the methods and limitations of its calculation in the cardiac catheterization laboratory.

Adolph Fick described the theoretical basis for cardiac output determination in man in 1870, but more than 60 years elapsed before clinical application became possible.¹ André Cournand² described the first set of cardiac output measurements in humans in 1945 and noted:

The practical difficulty preventing the ready application of this (i.e., Fick's) principle in human subjects has been that of obtaining reliable samples of average or mixed venous blood. With the development of the technique of catheterization of the right heart, this difficulty has been largely overcome, and the experience of the last 3 years has led to a procedure for determining cardiac output in man which can be used in almost all forms of disease or injury, with safety and without discomfort to the patient beyond that attendant upon the insertion of a needle in the femoral artery, and cutting down on a median basilic vein both under novocaine anesthesia.

Currently the determination of cardiac output and the calculation of the magnitude of an intracardiac shunt are routine and integral parts of cardiac catheterization. The performance of these measurements, their limitations, and their role in modern catheterization are discussed here.

MEASUREMENT OF CARDIAC OUTPUT

Cardiac output can be determined by one of several techniques: Fick (and "assumed Fick") methods, indicator-dilution method, thermodilution method, and angiography. Clinical catheterization laboratories and intensive care units rely primarily on the Fick and thermodilution methods. Cardiac output is expressed in liters/minute and is often corrected for patient size by dividing by the body surface area, which converts the measurement to the cardiac index in units of liters/minute/meter². The normal cardiac output at rest is 5 to 8 L/min, and the normal value for resting cardiac index is >2.4 L/min/m². With exercise, cardiac output increases substantially; elite athletes achieve cardiac outputs in excess of 30 L/min.

FICK METHOD FOR CARDIAC OUTPUT DETERMINATION

Adolph Fick described the theoretical basis for cardiac output determination nearly 150 years ago. Fick's principle states that the total uptake or release of a substance by an organ is the product of the *blood flow* to the organ and the *arteriovenous concentration difference* of that substance. Using the lungs as the organ and oxygen as the substance, blood flow to the lungs can be calculated by Fick's relationship, as follows:

$$\text{Pulmonary blood flow} = \frac{\text{Oxygen consumption}}{\text{Arteriovenous oxygen difference}}$$

In the absence of a shunt and because pulmonary blood flow equals systemic blood flow, the relationship can be used to calculate systemic cardiac output. The arteriovenous oxygen difference is the difference in oxygen content in the blood across the pulmonary circulation and is calculated as the arterial oxygen content minus the mixed venous oxygen content. Oxygen content is determined as the product of blood oxygen saturation, the hemoglobin concentration (in grams per deciliter), and the amount of oxygen carried per gram of hemoglobin (1.36 mL oxygen per gram of hemoglobin). This value is multiplied by 10 to correct the units (grams/deciliter to grams/liter). Thus Fick's formula for determining cardiac output is

$$\text{Cardiac output} = \frac{\text{Oxygen consumption}}{(\text{Arterial saturation} - \text{Mixed venous saturation}) \times (\text{Hgb}) \times 13.6}$$

Fick's principle states that the arteriovenous oxygen (AV-O₂) difference is inversely proportional to the cardiac output. A normal AV-O₂ difference is about 20 to 50 mL/L. At low cardiac outputs, there is greater extraction of oxygen

from the tissues, and the mixed venous saturation is low, resulting in a high AV-O₂ difference. In high-output states, diminished extraction leads to high mixed venous saturations and low AV-O₂ difference. Note that the mixed venous saturation alone is often used in clinical practices as a crude but rapid estimation of the cardiac output state, with a low mixed venous saturation indicating a low cardiac output and a high saturation indicating a high output (in the absence of a shunt). Furthermore, when cardiac output is low, the AV-O₂ difference is large, making it easier to measure and therefore making it a more accurate method at low cardiac outputs.

The important variables measured in the cardiac catheterization laboratory are the AV-O₂ content difference and the oxygen consumption. The AV-O₂ content difference is easily and accurately measured by simultaneously obtaining systemic arterial and mixed venous blood samples. Ideally, the arterial blood sample would be obtained from a pulmonary vein; however, in the absence of a right-to-left intracardiac shunt, the central aortic, femoral, or radial artery oxygen content closely approximates pulmonary vein oxygen content. In the absence of intracardiac shunting, the pulmonary artery is the most reliable site for obtaining mixed venous blood as opposed to other proximal sites such as vena cava, right atrium, or right ventricle, where there may be significant variation in blood oxygen content within the chamber. These blood samples are analyzed for the percentage of oxygen saturation. To determine oxygen content, the AV-O₂ difference is multiplied by the measured hemoglobin, which is then multiplied by the oxygen-binding coefficient (1.36 mL O₂/g of hemoglobin). The product is multiplied by 10 to convert units from grams/deciliter to grams/liter.

The other important measurement used to calculate the cardiac output by Fick's method, and a potential source of error, is oxygen consumption. At steady state, oxygen consumption is the rate at which oxygen is taken up by the blood from the lungs and should ideally be measured directly in the catheterization laboratory for each patient. Historically, oxygen consumption was directly measured using a Douglas Bag over a number of minutes, but this direct collection of expired air requires a steady-state environment, cooperation of the patient, and technical personnel who are familiar and experienced with the equipment and methodology. This method is time-consuming and cumbersome and has been largely abandoned in favor of the "assumed Fick" method, which is based on equations developed by LaFarge and Miettinen,³ as shown in Table 3.1. While relatively reliable and in widespread use today, the equations were originally developed using regression analysis of data from younger congenital heart disease patients with a mean age of 12 years and subsequently validated in broader patient cohorts. However, the observation, that oxygen consumption is known to vary during the cardiac catheterization procedure and may be significantly different from "assumed" values, has resulted in increasing interest in using more modern techniques to directly measure oxygen consumption in each patient concurrently with measurements of oxygen saturations. Cardiac output calculated by direct continuous oxygen consumption measurements can differ significantly from the calculated "assumed" values.^{4,5} Various modern, portable commercial systems can continuously determine oxygen consumption by measuring breath-by-breath oxygen levels using a facemask and can be used in the cardiac catheterization environment.⁴⁻⁶

Because of its simplicity, most catheterization laboratories and commercial hemodynamic systems still use an "assumed" value for oxygen consumption based on the patient's age, gender, and body surface area. The assumed value is typically 125 mL/min/m² for average individuals and 110 mL/min/m² for elderly patients. However, the obvious differences among patients who undergo cardiac catheterization introduce errors in cardiac output determinations as oxygen consumption varies significantly among adults at the time of cardiac catheterization, which introduces discrepancies between direct measurement of oxygen consumption and the assumed values,⁷⁻¹¹ particularly in patients with hypoperfusion or cardiogenic shock.¹²

Table 3.1 Formulas Used to Calculate Estimated Oxygen Consumption

AUTHORS, YEAR	NAME	EQUATION	DEVELOPMENTAL COHORT SUMMARY
Dehmer, Firth, and Hills, 1970 ⁸	Dehmer	$VO_2 = 125 \times BSA^a$	$n = 108$, mean age = 49 yrs (SD not provided), % male = 64, % cardiomyopathy = 9.0
LaFarge and Miettinen, 1970 ⁷	LaFarge	$VO_2 = 138.1 - (11.49 \times \log \text{age}) + (0.378 \times HR) \times BSA^a$ for males; $VC_2 = 138.1 - (17.04 \times \log \text{age}) + (0.378 \times HR) \times BSA$ for females	$n = 879$, mean age not provided, between 3 and 40 yrs, % male = 59, % cardiomyopathy not provided
Bergstra, van Dijk, Hillege, Lie and Mook, 1995 ⁹	Bergstra	$VO_2 = 157.3 \times BSA^a + 10 - (10.5 \times \log \text{age}) + 4.8$ for males; $VO_2 = 157.3 \times BSA^a - (10.5 \times \log \text{age}) + 4.8$ for females	$n = 250$, mean age 34.6 ± 22.7 yrs, % male = 57, % cardiomyopathy not provided

^aBSA calculated using the formula of Dubois¹⁶ with $BSA (m^2) = 0.007184 \times \text{weight (kg)}^{0.425} \times \text{height (cm)}^{0.725}$.
BSA, Body surface area; SD, standard deviation; VO_2 , oxygen consumption.

Although the use of “assumed” rather than directly measured O₂ consumption is the major source of error in the Fick method, additional errors may be introduced because of improper collection of the mixed venous or arterial blood samples. Ideally, these blood samples should be collected simultaneously in a steady-state environment. Errors may be present if, for instance, arterial blood is sampled early in the case, and mixed venous blood is sampled at a later time in the procedure, when the patient may become oversedated, thereby causing the patient to hypoventilate and resulting in the lowering of the mixed venous oxygen content, falsely implying a lower cardiac output. In addition, care should be taken to ensure that the mixed venous sample is obtained from the pulmonary artery and that the pulmonary artery catheter has not migrated to a “wedge” position that will falsely elevate the oxygen saturation. When the Fick method is performed correctly, the total error in the determination of the cardiac output is about 10%.¹³ Fig. 3.1 shows an example of Fick’s method for cardiac output calculation.

INDICATOR-DILUTION METHOD FOR CARDIAC OUTPUT DETERMINATION

Also based on Fick’s principle, the cardiac output can be derived by studying the flow characteristics of an indicator substance. On injection of a substance into the circulation, the rate at which the indicator appears and disappears from a downstream point correlates directly with the cardiac output. For example, if the cardiac output is high, the indicator will rapidly appear and quickly wash out; if the cardiac output is low, the indicator will require a longer time to achieve its maximal concentration and a longer time to wash out. Therefore the area under the time-concentration curve is related inversely to the cardiac output. The Stewart-Hamilton formula for the calculation of cardiac output takes into account the mean concentration of the indicator during the first passage and the duration of the extrapolated curve:

Cardiac output (L/min) = $\frac{\text{Amount of indicator in mg} \times 60 \text{ s/min}}{\text{Concentration of indicator (mg/mL)} \times \text{curve duration (s)}}$

Performance of this technique involves injection of a bolus of indicator into a systemic vein, the pulmonary artery, or the left atrium. The indicator mixes with the blood, and its concentration is measured continuously as a function of time at a sampling site (generally, the aorta, radial, or femoral artery). Numerous indicators have been used, with indocyanine green dye representing the most commonly used indicator. The normal curve generated with this method has an initial rapid upstroke followed by a slower downstroke and continued appearance due to the recirculation of the tracer. This recirculation creates some uncertainty at the tail end of the curve, and extrapolation of the curve is necessary to correct this distortion. The downslope of the primary curve is projected to the baseline to exclude indicator recirculation. Planimetry of the area under the curve yields the cardiac output.

Body surface area	2.0 m ²
Hemoglobin	15 g/dL
Mixed venous saturation	60%
Aortic saturation	99%
Oxygen consumption (assumed)	125 mL/min per m ²

Fick’s formula:

Cardiac output (L/min) = $\frac{\text{O}_2 \text{ consumption}}{\text{AV-O}_2 \text{ content difference}}$

Where:

AV-O₂ content difference = (Arterial – mixed venous saturation)(Hgb)(13.6)

Cardiac output (L/min) = $\frac{(125 \text{ mL/min per m}^2) \times (2.0 \text{ m}^2)}{(0.99 - 0.60) \times (15)(13.6)}$

Cardiac output = $\frac{250}{79.56} = 3.14 \text{ L/min}$

Fig. 3.1 Example of the Fick method used to calculate cardiac output in a 61-year-old male with shortness of breath. Note how the assumed O₂ consumption is multiplied by the body surface area. AV-O₂, Arteriovenous oxygen.

This method has several limitations. Indocyanine green dye is unstable over time and can be affected by light. The indicator must mix well with blood before reaching the distal sampling site, and it must have an exponential decay over time so that extrapolation of the time/concentration curve can be accurately performed. This technique is not accurate in the presence of irregular rhythms, valvular regurgitation, or intracardiac shunts. Importantly, this method is inaccurate in low output states in which the wash out of the indicator is prolonged. In these cases, recirculation of the indicator begins well before an adequate decline in the indicator curve occurs, distorting the downslope of the curve before it reaches baseline and preventing correction for recirculation. When indicator-dilution measurements are compared with the Fick method, the disparity between the two measurements is increased in patients with low cardiac output and those with aortic and mitral regurgitation. The disparity between the Fick method and indicator-dilution measurements is greater than the disparity between Fick and thermodilution measurements.¹⁴ This technique has essentially been abandoned by clinicians and is primarily of historical interest.

THERMODILUTION METHOD FOR CARDIAC OUTPUT DETERMINATION

Fegler¹⁵ described the thermodilution method in 1954. This method is the easiest to perform and the most widely used method to measure cardiac output in catheterization laboratories and critical care units.

The thermodilution technique is a variation of the indicator-dilution technique, using blood temperature as the indicator. Saline at a known temperature is injected into the right atrium from the proximal port of a Swan-Ganz catheter. The saline mixes with the blood and lowers its temperature. The temperature of blood is measured in the pulmonary artery by a thermistor mounted on the distal tip of the catheter. The thermistor is a variable resistor in which the resistance is proportional to the temperature. As the resistance changes, a change in voltage occurs. The measured change in voltage over time generates a temperature curve that is related to the cardiac output. Similar to the indicator-dilution method, cardiac output is inversely related to the area under the time-temperature curve. If the area under the curve is small, this means the temperature equilibrates rapidly with the ambient body temperature, indicating a high cardiac output. Conversely, if the area under the curve is large, it takes longer for the blood temperature to reach ambient body temperature, implying low cardiac output. The cardiac output is calculated using an equation that takes into account the temperature and specific gravity of the injectate and blood and the injectate volume.

The thermodilution method is preferable to the indicator-dilution method because right-sided injection and right-sided sampling of the cold saline yield a curve that is less subject to recirculation-induced distortion than right-sided injection and left-sided sampling of indocyanine dye. Examples of a modern, computer-based system to measure cardiac output using the thermodilution technique are shown in Figs. 3.2 and 3.3. Note that the temperature plotted on the y-axis decreases as the saline bolus passes the thermistor on the pulmonary artery catheter. The curve of a patient with normal output shows a rapid drop in temperature, whereas patients with low output have a much slower drop in temperature.

Thermodilution cardiac output is very simple to perform, but there are several potential sources of error. After attaching the Swan-Ganz catheter to the measuring computer, the operator rapidly injects a 10-mL bolus of saline into the right atrium via the proximal port of the catheter (volume may vary depending on the catheter size and associated computer settings). No infusions should be entering the right atrium from large peripheral or central lines. Ideally, the

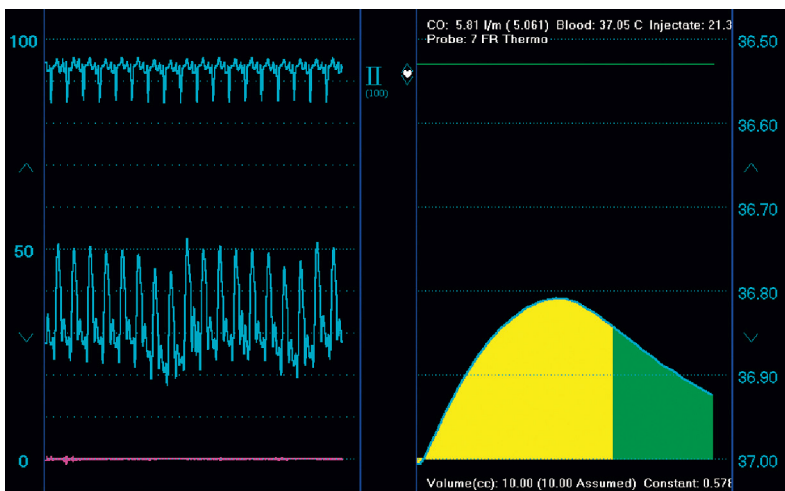


Fig. 3.2 Thermodilution curves were obtained from a patient with a normal cardiac output. The temperature is plotted on the y-axis. In this case, the cardiac output measured 5.81 L/min.

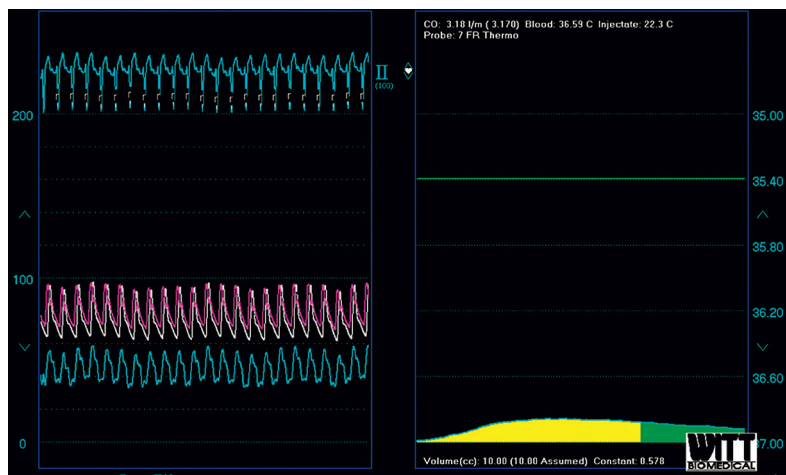


Fig. 3.3 Thermomodulation curves were obtained in a patient with a profoundly depressed cardiac output from cardiogenic shock due to pump failure. Note the flattened shape of the curve compared to the normal curves obtained in Fig 3.2. The decrease in temperature over time is minimal because of the slow flow. In this case, the cardiac output measured 3.18 L/min.

same amount of injectate at the same temperature should be used each time. Therefore the operator must take great care to rapidly deliver each bolus, at a constant rate and with minimal contact with the syringe barrel to prevent warming and the introduction of saline with widely varying temperatures with each injection. Using iced saline does not improve accuracy. The use of a dual thermistor catheter (located at both proximal and distal ports) improves the precision and accuracy of this technique.¹⁶ While measuring cardiac output, the patient should be resting quietly without talking, laughing, or coughing because these will vary the thoracic temperature and affect the measurement. The distal catheter tip should be in a stable position in the pulmonary artery, with avoidance of the “wedge” position. All redundancy or coiling of the catheter in the right atrium or right ventricle should be removed. Routinely, three to five cardiac output measurements are obtained and averaged. If wide variation in measured values occurs, for example, in a patient with atrial fibrillation and irregular heart rate, additional samples should be obtained and averaged.

The thermodilution method is inaccurate in the presence of intracardiac shunts, low flow states, marked respiratory variation, or cardiac arrhythmia. This method is likely inaccurate in severe tricuspid regurgitation because of the unpredictable mixing and loss of heat with regurgitation; however, studies have shown conflicting results regarding its accuracy in this condition when compared to other techniques.^{17–19} In low cardiac output states, the thermodilution method overestimates cardiac output when compared to the Fick method. This overestimation is greatest (average difference of 35%) in patients with Fick cardiac outputs <2.5 L/min.²⁰

ANGIOGRAPHIC METHOD FOR CARDIAC OUTPUT DETERMINATION

Cardiac output equals the product of stroke volume and heart rate. Left ventriculography can provide an estimation of stroke volume by tracing the contours of a high-quality ventriculogram and calculating the end-diastolic and end-systolic volumes. Stroke volume represents the difference between these values and is simply multiplied by the heart rate at the time of the ventriculogram. Obviously, this method is fraught with inaccuracies because of the inherent errors of calibrating chamber volumes by ventriculography and is particularly inaccurate in patients with valvular regurgitation and irregular heart rhythms, such as atrial fibrillation. This method is rarely used clinically.

CALCULATION OF SHUNTS

In the absence of a shunt, blood flow through the pulmonary and systemic circulations is equal. The presence of abnormal communication between intracardiac chambers results in shunting of blood either from the systemic to the pulmonary circulation (left-to-right shunt), the pulmonary-to-the-systemic circulation (right-to-left shunt), or in both directions (bidirectional shunt). An intracardiac shunt is most commonly due to congenital heart disease, such as an atrial septal defect (ASD), ventricular septal defect (VSD), or patent ductus arteriosus (PDA). Shunts can also be iatrogenic or acquired conditions—for example, as from procedures involving crossing the interatrial septum with larger catheters (as during mitral valve clip placement or mitral balloon valvuloplasty), a postsurgical fistula, a postmyocardial infarction rupture of the interventricular septum, or a complication of endocarditis.

SCREENING FOR INTRACARDIAC SHUNTS AND PERFORMANCE OF A “SHUNT RUN”

Routine right-heart catheterization procedures should screen for the presence of a left-to-right shunt by measuring the oxygen saturation of blood from the pulmonary artery. Normally, pulmonary artery blood saturation does not exceed 75%; higher values indicate either a high cardiac output or a left-to-right shunt. Some variability in oxygen content in healthy people is observed if multiple samples are obtained from any particular chamber. For instance, the right atrium receives blood from the superior vena cava (SVC), the inferior vena cava (IVC), and the coronary sinus, all of which have varying degrees of oxygen saturation. Similarly, coronary venous blood, with its very low oxygen content, may directly enter the right ventricle through the Thebesian vessels. Therefore oxygen saturation measured in blood samples obtained from the right atrium or right ventricle will vary depending on where blood is collected in this chamber relative to these sources. A *significant step-up* is defined as one that exceeds the normal variability. Generally accepted criteria for a significant shunt are a mean absolute difference of $>7\%$ at the atrial level and a mean absolute difference of $>5\%$ at the ventricular and great vessel levels.²¹

A better method of screening for a left-to-right shunt is to measure oxygen saturation in blood samples obtained from the SVC and the pulmonary artery. Again, these should be within 7% of each other; a variance of more than this indicates a left-to-right shunt, and a difference in oxygen saturation between these two samples of $>9\%$ provides excellent sensitivity, specificity, and predictive accuracy for identifying a large left-to-right shunt.²² A right-to-left shunt is present when oxygen desaturation is discovered in arterial blood that does not correct with the administration of 100% inhaled oxygen.

A full oxygen saturation run should be performed when an intracardiac shunt is suggested by the shunt screen or in cases of known or suspected shunts (often observed by echocardiography or other cardiac imaging). A complete oximetry run involves obtaining samples from multiple levels within the heart (Fig. 3.4). These include samples obtained from the left and right pulmonary arteries; the main pulmonary artery; the midcavity of the right ventricle and right ventricular outflow and inflow tracts; the low, middle, and high right atrium; the low and high SVC; the low and high IVC; the left ventricle; and the distal aorta. The IVC saturation varies depending on where the sample is obtained, and the sampling site should be at the level of the diaphragm to ensure that hepatic venous blood is taken into account. If an ASD is discovered, the catheter should be placed across the atrial septum, and a blood sample should be taken from both the left atrium and pulmonary veins. The typical catheter position across the atrial septum is confirmed by the catheter lying across the midline in the anteroposterior projection. Further confirmation can be obtained by the presence of a left atrial waveform and by obtaining an oxygen saturation exceeding 90%. These multiple samples allow a precise determination of the location of the anatomic shunt by identifying the site of the oxygen step-up within the heart. For example, if there is an isolated step-up in the low SVC or in the high right atrium, anomalous pulmonary venous drainage commonly associated with a sinus venosus ASD can be suspected.

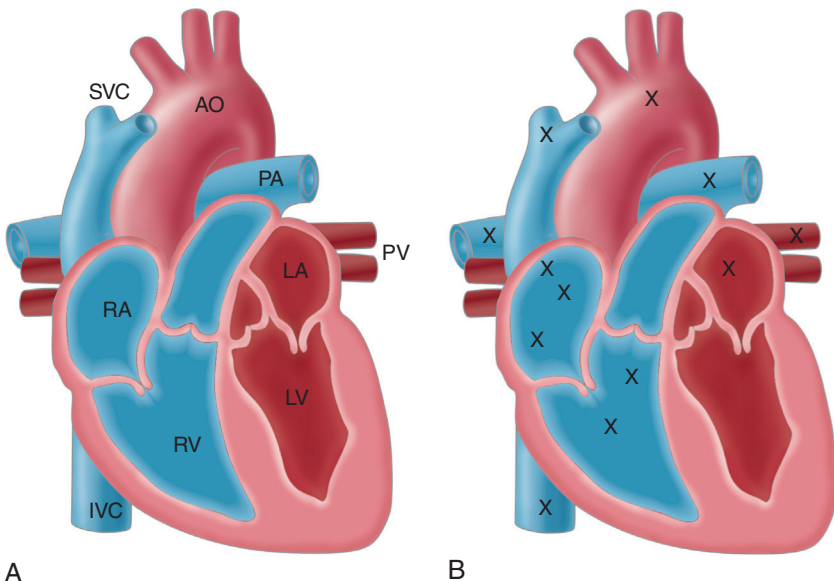


Fig. 3.4 Locations for blood sampling when performing a complete saturation run. (A) Various cardiac chambers. (B) Optimal sites are marked by an x. AO, aorta; IVC, inferior vena cava; LA, left atrium; LV, left ventricle; PA, pulmonary artery; PV, pulmonary vein; RA, right atrium; RV, right ventricle; SVC, superior vena cava.

Importantly, a shunt run should be performed in an expeditious manner, with all samples obtained in prompt succession to ensure maintenance of a steady state; the entire oximetry run should take less than 7 min. Samples should be acquired using an end-hole catheter with its position confirmed by pressure waveforms and fluoroscopy with the patient breathing room air or ventilated with <30% oxygen.

CALCULATION OF SHUNT SIZE

The magnitude of a shunt can be expressed in terms of either absolute blood flow in liters/minute or, more commonly, a ratio of pulmonary blood flow to systemic blood flow. A left-to-right shunt will cause an increase in the pulmonary blood flow relative to the systemic blood flow, whereas a right-to-left shunt causes increased systemic blood flow relative to the pulmonary blood flow. The shunt-size calculation is based on the individual determination of systemic and pulmonary blood flow using Fick's principle, as described earlier. Therefore systemic blood flow, or Q_s , is determined, as follows:

$$Q_s = \frac{\text{Oxygen consumption}}{(\text{Arteriovenous oxygen content difference across the body})}$$

The arteriovenous oxygen content difference across the body is first determined by measuring the oxygen saturation of aortic blood and subtracting the oxygen saturation of mixed venous blood. This value is then multiplied by the hemoglobin concentration, the amount of oxygen carried per gram of hemoglobin (1.36), and the number 10 to correct the units.

Importantly, in the presence of a left-to-right shunt, pulmonary artery blood *does not* represent mixed venous blood. The appropriate cardiac chamber that represents mixed venous blood depends on the location of the shunt. One method is to simply choose the chamber proximal to the shunt. For example, the right atrium can be sampled if there is a VSD. This becomes more difficult if the shunt is located at the level of the right atrium (i.e., an ASD or anomalous pulmonary venous return). In these situations, both the SVC and the IVC sources are taken into account, as described by the Flamm²³ formula:

$$\text{Mixed venous oxygen content} = \frac{3(\text{SV CO}_2 \text{ content}) + 1(\text{IVCO}_2 \text{ content})}{4}$$

Similar to systemic blood flow, pulmonary blood flow, or Q_p , can be determined by the relationship:

$$Q_p = \frac{\text{Oxygen consumption}}{(\text{Arteriovenous oxygen content difference across the lung})}$$

The arteriovenous oxygen content difference across the lung is determined by measuring the oxygen saturation of blood in the pulmonary vein and subtracting the oxygen saturation of pulmonary artery blood. This value is then multiplied by the hemoglobin concentration, the amount of oxygen carried per gram of hemoglobin (1.36), and the number 10 to correct the units.

In the case of a left-to-right shunt, once pulmonary and systemic blood flows are calculated, shunt flow can be determined by the relationship, as follows:

$$\text{Pulmonary blood flow (} Q_p \text{)} = \text{Systemic blood flow (} Q_s \text{)} + \text{left-to-right shunt flow}$$

In the case of right-to-left shunt:

$$\text{Systemic blood flow (} Q_s \text{)} = \text{Pulmonary blood flow (} Q_p \text{)} + \text{right-to-left shunt flow}$$

The determination of the absolute flow rates for pulmonary and systemic blood flow is important for the calculation of pulmonary vascular and systemic vascular resistances. However, in most cases, the ratio Q_p/Q_s is used to determine the significance of the shunt. This ratio can be easily calculated by combining the equations for systemic and pulmonary flow equations and canceling the common terms (oxygen consumption, hemoglobin, and oxygen carrying capacity), as follows:

$$\frac{Q_p = \text{O}_2 \text{ consumption} / (\text{pulmonary venous oxygen content} - \text{pulmonary arterial oxygen content})}{Q_s = \text{O}_2 \text{ consumption} / (\text{arterial oxygen content} - \text{mixed venous oxygen content})}$$

Or, more simply,

$$\frac{Q_p = \text{Systemic arterial saturation} - \text{mixed venous saturation}}{Q_s = \text{Pulmonary venous saturation} - \text{pulmonary arterial saturation}}$$

Hemoglobin (Hgb)	12 g/dL
Body surface area (BSA)	1.8 m ²
O ₂ consumption	125 mL/min per m ²
<u>Oximetry Data:</u>	
Superior vena cava (SVC)	69%
Inferior vena cava (IVC)	73%
Low right atrium	79%
Mid right atrium	91%
High right atrium	69%
Right ventricle	93%
Left atrium (LA)	98%
Pulmonary vein (PV)	98%
Pulmonary artery (PA)	90%
Aorta (AO)	98%

A

Step 1: Determine mixed venous saturation:

$$\text{Mixed venous (MV) saturation} = \frac{3(\text{SVC}) + (\text{IVC})}{4} = \frac{3(0.69) + (0.73)}{4} = 0.70$$

Step 2: Calculate pulmonary blood flow (Qp)

$$Q_p = \frac{\text{O}_2 \text{ consumption} \times \text{BSA}}{(\text{PV sat} - \text{PA sat})(\text{Hgb})(13.6)} = \frac{225}{(0.98 - 0.90)(12)(13.6)} = 17.2 \text{ L/min}$$

Step 3: Calculate systemic blood flow (Qs)

$$Q_s = \frac{\text{O}_2 \text{ consumption} \times \text{BSA}}{(\text{AO sat} - \text{MV sat})(\text{Hgb})(13.6)} = \frac{225}{(0.98 - 0.70)(12)(13.6)} = 4.9 \text{ L/min}$$

B

Step 4: Calculate shunt flow

$$\text{Shunt flow} = Q_p - Q_s = 17.2 - 4.9 = 12.3 \text{ L/min}$$

$$\text{Shunt ratio} = Q_p/Q_s = 17.2/4.9 = 3.5 \text{ to } 1$$

Step 5: Determine Qp/Qs using simplified formula

$$\frac{Q_p}{Q_s} = \frac{(\text{AO sat} - \text{MV sat})}{(\text{PV sat} - \text{PA sat})} = \frac{28}{8} = 3.5 \text{ to } 1$$

C

Fig. 3.5 Example of the calculation of the magnitude of a left-to-right shunt in a 42-year-old female with a secundum atrial septal defect. (A) The raw data for the calculations. (B) and (C) Calculations.

The minimal detectable shunt by the oxygen saturation method is a Q_p/Q_s ratio of 1.3 to 1. A Q_p/Q_s ratio between 1.0 and 1.5 indicates a small left-to-right shunt, and a $Q_p/Q_s > 2.0$ indicates a large left-to-right shunt. A $Q_p/Q_s < 1.0$ indicates a net right-to-left shunt. An example of a calculation of the size of a left-to-right shunt is shown in Fig. 3.5. Examples of conditions associated with left-to-right shunting include ASD, anomalous pulmonary venous return, VSDs,

and PDA among others. Examples of conditions with right-to-left shunting include tetralogy of Fallot, Eisenmenger ASD or VSD, pulmonic or tricuspid valve atresia, among others. Right-to-left shunting may be suspected in patients with cyanosis or arterial desaturation. It is important to understand that determining the site and magnitude of right-to-left shunting requires that blood samples be obtained from one or more pulmonary veins, the left atrium, the left ventricle, and the aorta, which is the most significant limitation to performing oximetry for such shunting in the cath lab. However, in cases where pulmonary venous oxygen saturation can be safely measured and is normal in a patient with cyanosis or hypoxemia, then right-to-left shunting can be suspected.

There are several limitations to the oximetric technique for quantifying a shunt. The patient should be in a steady state, and measurements from chamber to chamber should be made expeditiously. The measurement of oxygen saturation does not account for the dissolved form of oxygen. If, for example, a patient is mechanically ventilated with high fractions of inspired oxygen (>30%), a significant amount of oxygen may be present in dissolved form in the pulmonary venous sample, and saturation data may not provide accurate information regarding pulmonary blood flow for shunt calculation. The detection of a shunt by oximetry is dependent on the rate of systemic blood flow. For example, high systemic flow equalizes the arteriovenous oxygen difference across the systemic bed, leading to higher mixed venous oxygen saturation. In this case, even a small increase in right-heart oxygen measurements will indicate a significant left-to-right shunt. On the other hand, if systemic blood flow is reduced, the level of mixed venous oxygen saturation is low, and a larger increase must be detected to determine a significant left-to-right shunt.²⁴ Oximetrically derived Qp/Qs are subject to substantial inpatient variability.²⁰ Small differences in pulmonary arterial saturations, for example, produce a substantial difference in the arteriovenous oxygen content across the lungs, which, in turn, lead to a sizable difference in calculated pulmonary blood flows and Qp/Qs ratios. The variability in Qp/Qs is greatly diminished by taking multiple blood samples from each chamber and averaging them before calculating the Qp/Qs .²⁵ Finally, the degree of shunting may vary. For example, the volume of blood shunting with an ASD depends not only on the size of the defect but also on the compliance of the left and right ventricles and is therefore subject to sympathetic tone and preload and afterload conditions. Additionally, some shunting is physiological (thebesian and bronchial circulations).

INDICATOR-DILUTION METHOD OF SHUNT DETECTION AND CALCULATION

The indicator-dilution, or *dye curve*, method is a sensitive and accurate method for the detection and quantitation of intracardiac shunts, but this technique is time-consuming and cumbersome to perform and is no longer routinely used for this purpose. An indicator (indocyanine green dye) is injected into a proximal chamber and a sample is taken from a distal chamber. Using a densitometer, the density of the dye is displayed over time. To detect a left-to-right shunt, dye is injected into the pulmonary artery, and sampling is performed in a systemic artery. The presence of a shunt is indicated by the early recirculation of the dye noted on the downslope of the curve as a secondary rise in concentration. To detect a right-to-left shunt, dye is injected into the right side of the heart proximal to the location of the suspected shunt, and blood samples are obtained from a systemic artery. A right-to-left shunt is revealed by a distinct, early peak present on the upslope of the curve. The indicator-dilution method is more sensitive than the oximetric method in the detection of small shunts, but it cannot localize the shunt. Overall, the agreement is favorable between the two methods, especially concerning the pulmonary-to-systemic flow ratios.²⁶

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